XIYUAN (TOMMY) GUO

B.A. Candidate in Computer Science, University of Virginia
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Theoretical quantum information; quantum error correction and quantum LDPC decoding; constrained optimization; computational imaging.

EDUCATION

University of Virginia — Expected May 2027
B.A. in Computer Science — *Charlottesville, VA*

- **GPA:** 3.73/4.00
- **Relevant Coursework:** Graduate Quantum Computing (PHYS 5880, Prof. I. Klich; in progress).
- **Independent Study:** Quantum information (Preskill lectures; Solovay–Kitaev theorem and approximation bounds); quantum mechanics (Griffiths, Sakurai).

RESEARCH EXPERIENCE

Research Assistant — Feb 2026 – Present
Yale University (Prof. Mengxia Liu's Group) — *New Haven, CT*

- Built a joint spectro-polarimetric forward model that recovers a 324-channel datacube (spatial $\times$ 81 wavelength bands $\times$ 4 Stokes parameters) from a single multiplexed lensless speckle image.
- Formulated the reconstruction as a conic-constrained inverse problem, enforcing a Lorentz-cone constraint on the Stokes vector ($S_0 \geq \|S_{1:3}\|$) to guarantee physically admissible polarization states.
- Implemented a consensus ADMM solver in PyTorch with an exact joint proximal operator encoding the physical constraints, L_1 spectral sparsity, and Sherman–Morrison linear solves; runs on GPU via SLURM.
- Calibrated PSF libraries with Gram–Schmidt orthogonalization to remove channel cross-talk and performed spectral unmixing via non-negative least squares.

Research Assistant — Apr 2026 – Present
UVA BAL Lab (Prof. Xin Cynthia Tong's Group) — *Charlottesville, VA*

- Developed PRISM, a constrained-projection framework for multivariate longitudinal missing-data imputation that projects initial imputations toward the model-implied covariance structure, correcting the variance attenuation of standard conditional-expectation methods.
- Built the solver in C++17 (RcppArmadillo) with projected gradient descent, Armijo line search, exact mean-constraint projection, and KKT optimality checks.
- Tested the framework on multi-qubit state tomography: the random-forest estimator did not transfer, while the constrained-projection layer did, demonstrating the value of the projection structure.

Research Assistant — Oct 2025 – Present
UVA (Prof. Liqin Ke's Group) — *Charlottesville, VA*

- Studied transition-metal magnetic anisotropy and its relationship to structural motifs in crystal lattices with density functional theory (DFT).
- Built interactive 3D atomic-structure visualizations (Three.js, JSmol) for real-time structural inspection.
- Developed a shared ML pipeline (Go, Next.js) for collaborators to run predictive models on magnetic-property datasets.
- Built the schema, API, and web interface for RareEarth DB, a rare-earth analogue of the Materials Project; the database curates 4,648 anisotropy measurements linked to ICSD/Pearson structures, with Fireworks-orchestrated DFT workflows on UVA HPC. DOE funding proposal in preparation.

Research Intern May 2026 – Aug 2026
UVA Biocomplexity Institute (Prof. Abhijin Adiga's Group) *Charlottesville, VA*

- Engineered a Julia simulation engine for Independent Cascade diffusion and trained PyTorch Geometric GCNs to estimate total epidemic cascade size from sparse, partially observed network infections.
- Developed an agentic workflow engine (Pydantic AI, NetworkX) that translates natural-language queries into verified execution DAGs for network-analysis pipelines, with deterministic validation and graph deduplication cutting redundant computation by $> 40\%$.

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

- Guo, X., [Co-authors], Ke, L. "Transition-metal magnetic anisotropy and its relationship to structural motifs." *[Manuscript in preparation]*.
- Guo, X., [Co-authors], Liu, M. "Joint Spectro-Polarimetric Lensless Reconstruction via Conic-Constrained ADMM." *[Manuscript in preparation]*. Code available on request.
- Guo, X., [Co-authors], Tong, X.C. "PRISM: Lagrangian Projection Engines for Multivariate Longitudinal Missing Data Imputation." *[Manuscript in preparation]*. Code: github.com/xguot/prism.

OPEN SOURCE SOFTWARE

qudec: Quantum LDPC Decoding Toolkit
Open-source project *github.com/xguot/qudec*

- Implemented LP-based and batched PyTorch ADMM decoders for quantum LDPC codes, including exact parity-polytope projection via its two-slice representation.
- Verified decoder correctness against brute-force maximum-likelihood decoding on small codes, then benchmarked against BP+OSD on bivariate bicycle codes: $[[144, 12, 12]]$ under code-capacity noise, $[[72, 12, 6]]$ under phenomenological noise.
- Tuned the ADMM penalty ($\rho = 4$ plus an OSD sweep) to close the gap with BP+OSD, normalized LLR penalty weights to prevent trivial fixed points, and characterized BP failure modes on degenerate codes.
- **stim:** Authored pull request #1099 to Stim, Google's open-source quantum stabilizer circuit simulator; accepted by the maintainers.
- **sanjaya:** Academic literature extraction pipeline (OpenAlex API, Scrapy, Playwright).
- **difi:** Go-based Git diff TUI; Homebrew Core, 365+ GitHub stars, #5 on Hacker News.

HONORS, AWARDS, AND GRANTS

- **Double Hoo Research Grant**, University of Virginia ($5,000) *2026*
- **Dean's List of Distinguished Students**, University of Virginia
- **Silver Division**, USA Computing Olympiad (USACO)

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Julia, Rust, Go, TypeScript, R, SQL, Bash, LaTeX. **Scientific computing:** PyTorch/CUDA, JAX, PyTorch Geometric, scikit-learn, pandas, RcppArmadillo. **Systems & HPC:** Linux, Docker, K3s Kubernetes, Slurm HPC, Git. **Math & Physics:** Quantum mechanics, quantum information, convex optimization, linear algebra, matrix calculus.

COMMUNITY SERVICE

Volunteer Firefighter 2022 – 2024
Plainfield Fire Department *Plainfield*